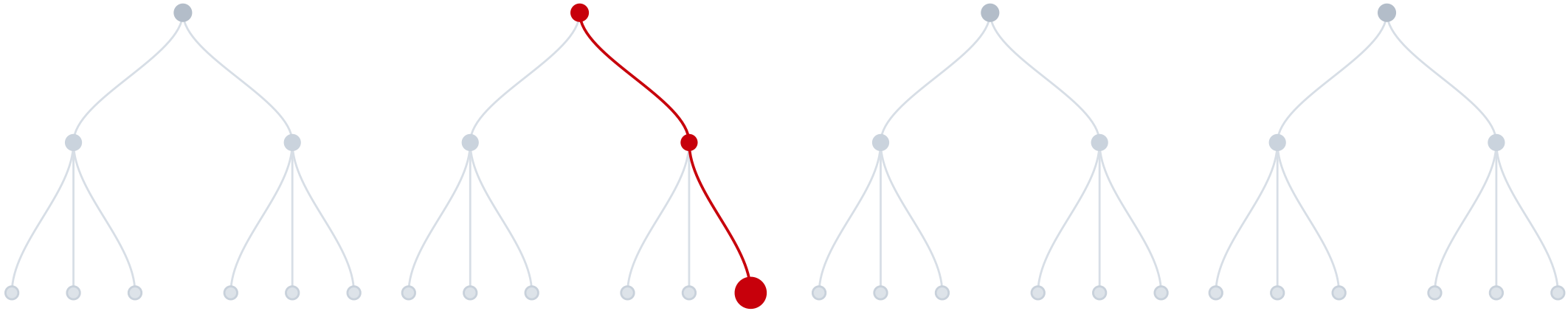
Technical Solution: Hierarchical Discovery Pipeline

A search for a better mathematical solution

In this Erdős example, lower scores are better.



Starting candidate (parent)

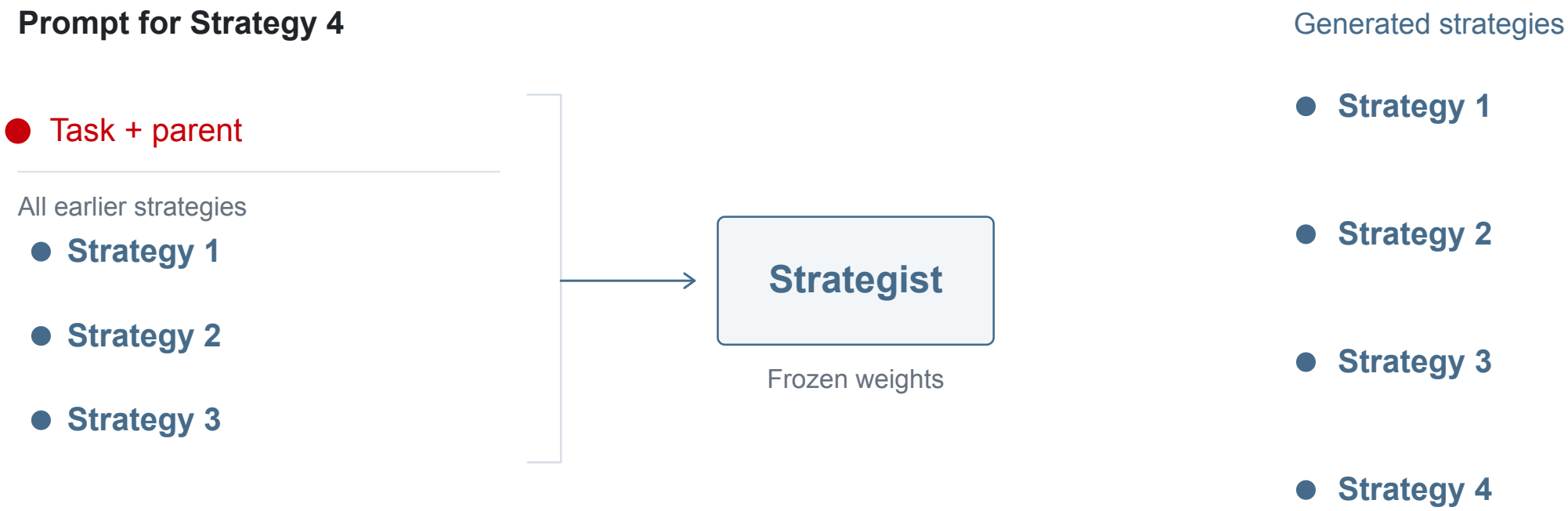
Score: 0.509541

Each dot is a saved candidate. Each link records where it came from.

A candidate is a mathematical construction produced by a program.

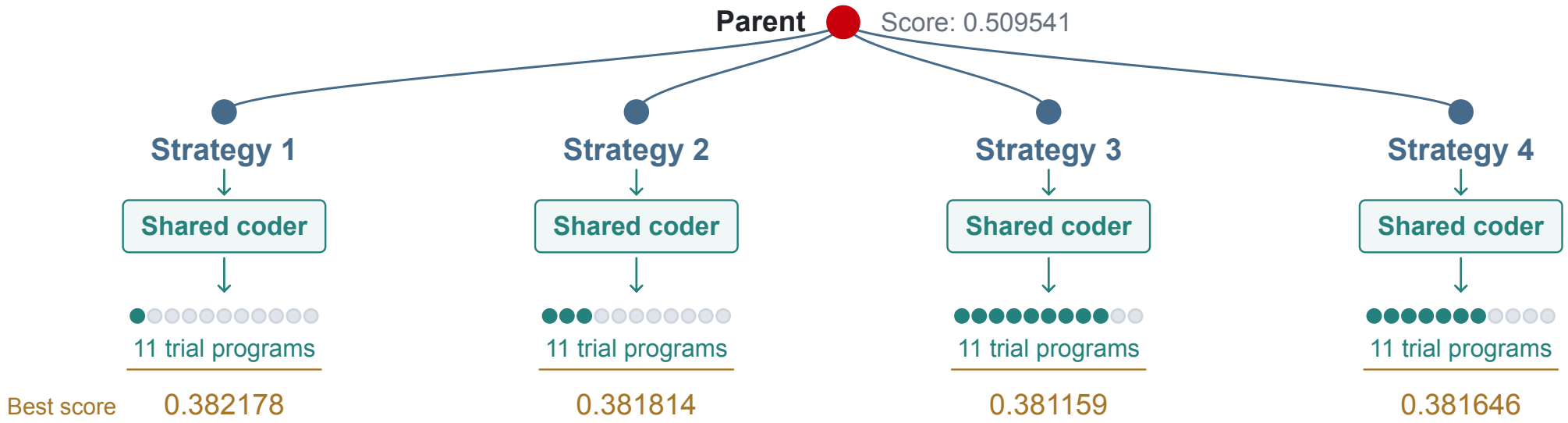
Technical Solution: Hierarchical Discovery Pipeline

Each strategy uses the parent and every earlier strategy



Technical Solution: Hierarchical Discovery Pipeline

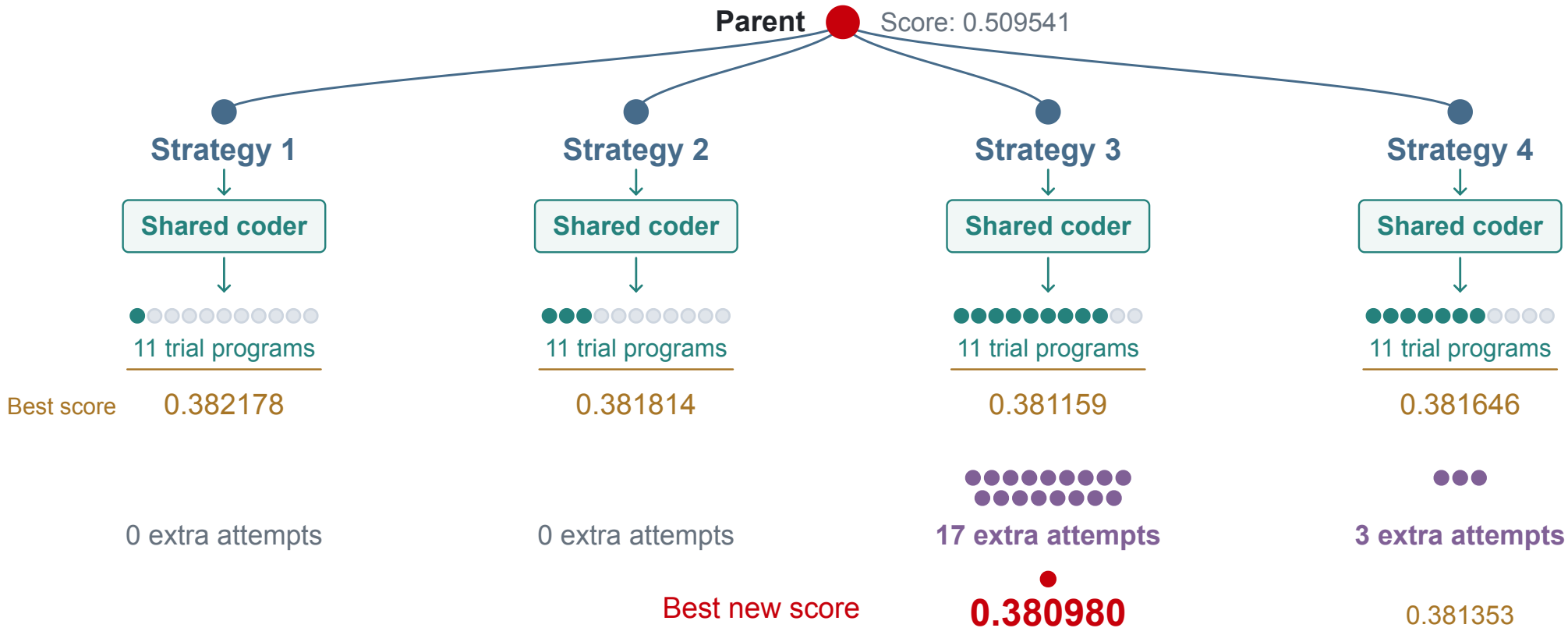
The coder turns each strategy into testable candidates



The evaluator runs the code, checks its output, and assigns a score.

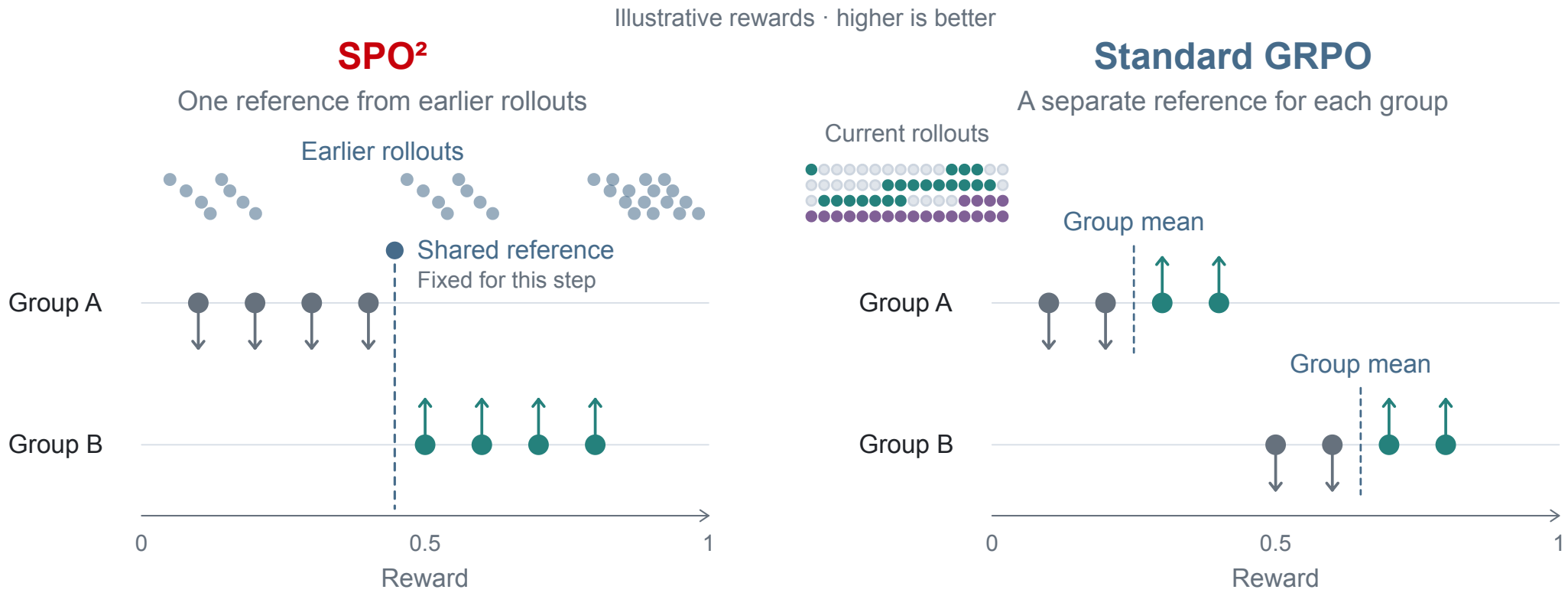
Technical Solution: Hierarchical Discovery Pipeline

The bandit allocates more attempts using pilot results



Technical Solution: Hierarchical Discovery Pipeline

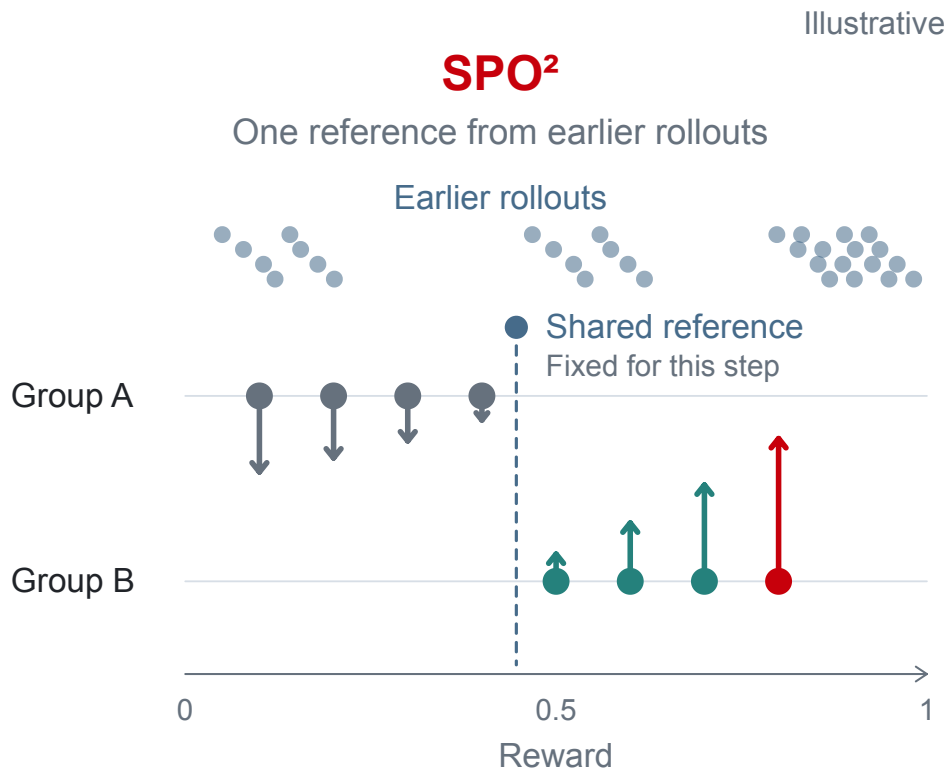
Training signals for the same eight rewards



Direction only: ↑ reinforce ↓ discourage

Technical Solution: Hierarchical Discovery Pipeline

One SPO² update from all evaluated rollouts



Coding model

Base weights

Frozen

LoRA Updated

One optimizer update

The strategist stays frozen

SPO² arrow length shows the learning weight; every rollout contributes.

Technical Solution: Hierarchical Discovery Pipeline

The next step uses a saved child and the updated coder

